

Title

E-Health Supported Lifestyle Modification and Pregnancy Outcomes: A Comparative Effectiveness Study

Abstract

We conducted a multisite randomized trial of a digital lifestyle intervention designed to improve metabolic health and outcomes in subsequent pregnancies. Women recruited in the immediate postpartum period were randomized at 6 weeks to receive either structured digital coaching combined with mobile health monitoring ($n = 187$) or standard postpartum care ($n = 191$). The intervention was delivered across the interpregnancy period and during the subsequent pregnancy. The primary composite outcome (any occurrence of pregnancy-induced hypertension, gestational diabetes mellitus, caesarean delivery, or delivery of a large-for-gestational-age infant) occurred in 34% of the intervention group compared to 47% in the control group (adjusted odds ratio 0.59, 95% CI 0.41-0.85). Mean gestational weight gain was significantly lower in the intervention group (8.1 kg vs 10.4 kg, $p = 0.002$). Participant engagement with the application and coaching sessions was variable; approximately 71% completed at least 75% of coaching sessions. Weight loss in the interpregnancy period predicted better subsequent pregnancy outcomes. The intervention was generally well-tolerated with high satisfaction ratings (85% reported satisfaction or high satisfaction). This multisite study provides evidence for digital lifestyle interventions in preventing adverse pregnancy outcomes.